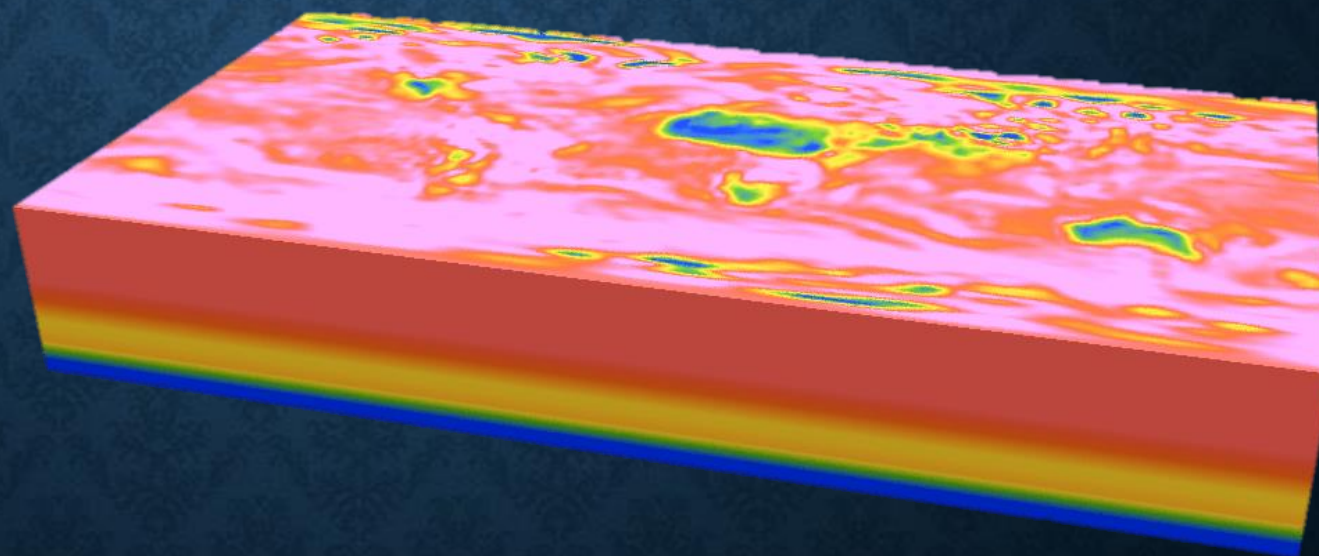


Visualization Tools

Demonstration of powerful tools for
visualization of small and large datasets



Center for Research Computing/OIT

Bruno Sousa

Uilvim E. G. Franco

Geospatial Analytics & Data Visualization Specialists

CONTENT / AGENDA

- **Introduction**
 - CRC Services
 - Spatial Studies Lab
- **Mapping & Geospatial APIs**
 - ArcGIS JavaScript API
 - **Mapbox GL JS**
 - Leaflet JS
 - Deck.gl
- **3D Graphics Libraries**
 - Three.js
 - Babylon.js
- **Data Visualization Libraries**
 - Chart.js
 - D3.js
- **ArcGIS Online**
 - First map

CENTER FOR RESEARCH COMPUTING SERVICES



High-Performance Computing: Parallelization, code optimization, GPUs for machine learning, HPC application support



Cloud for Research:
Virtual machines on demand, hybrid cloud solutions



Research Data Storage: Managing, securing, and hosting research data, moving large datasets, sharing data with colleagues



Research Systems Management:
Procuring, deploying, and managing research computing resources for your lab



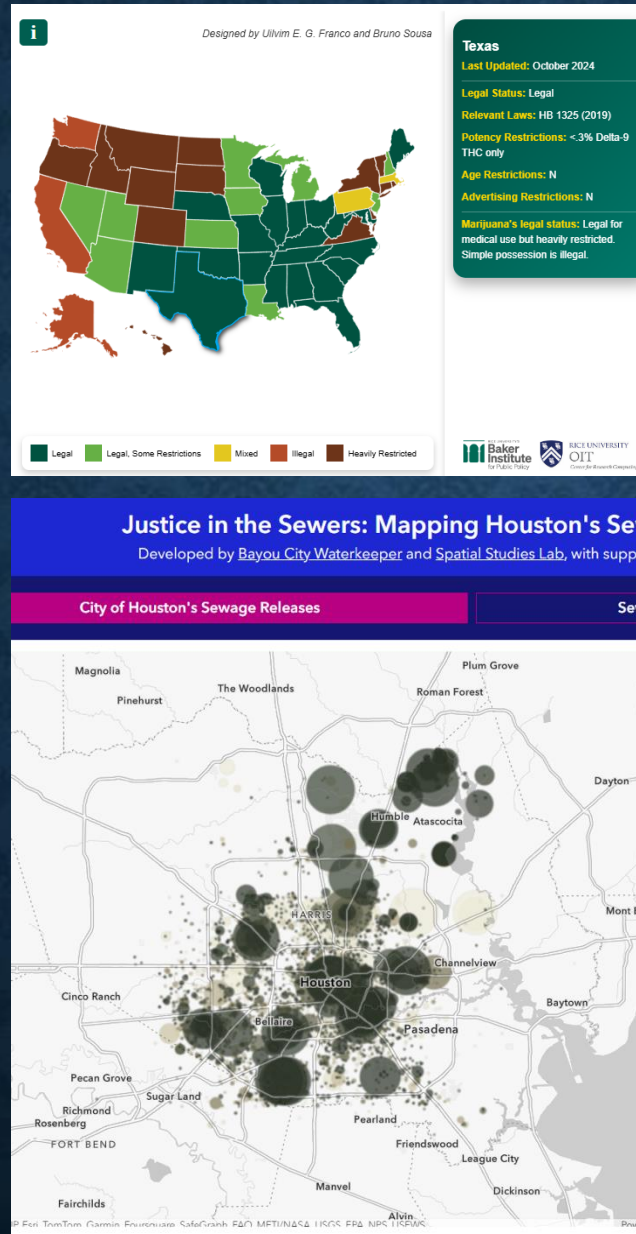
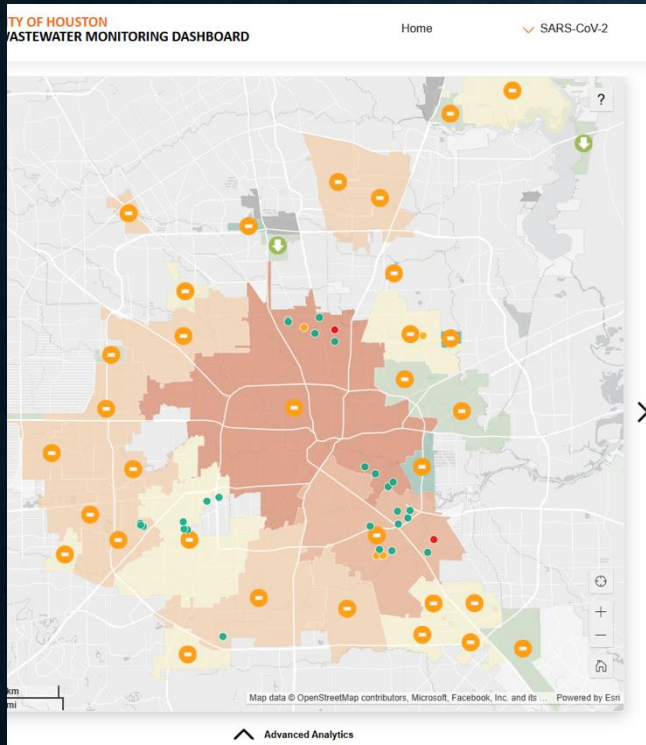
Data Analysis, Mapping, and Visualization:
Data cleaning and preparation, geospatial data collection, GIS mapping, spatial analysis, data visualization



Facilitation and Training:
Training for Rice's shared research computing resources, consultation for computing solutions and workflows, workshops to develop core computing skills and concepts

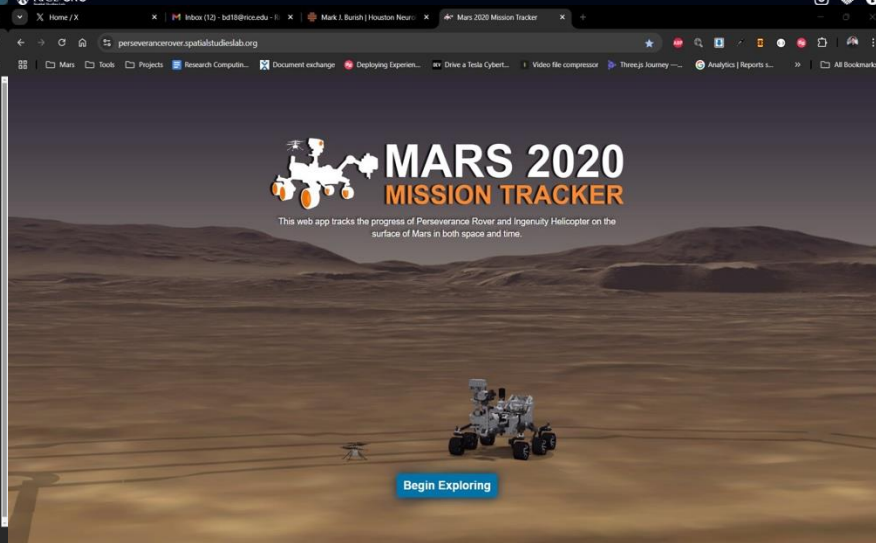
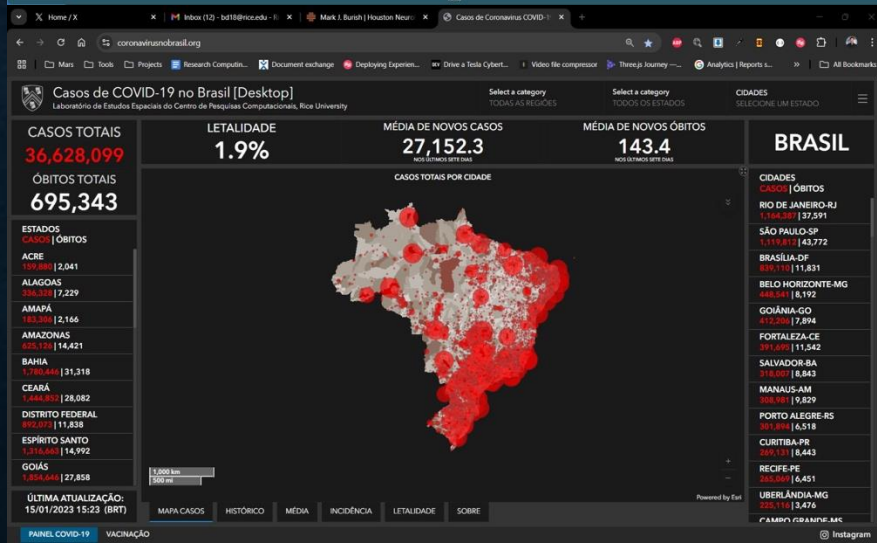
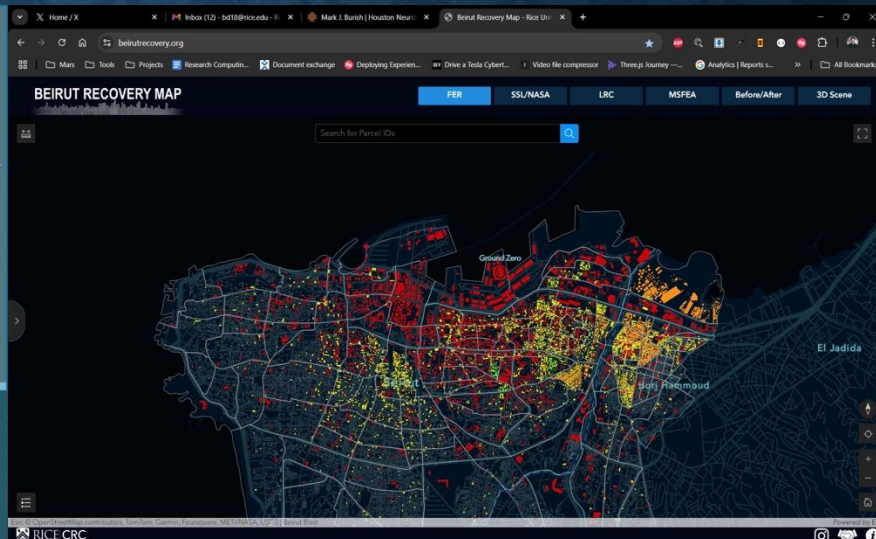
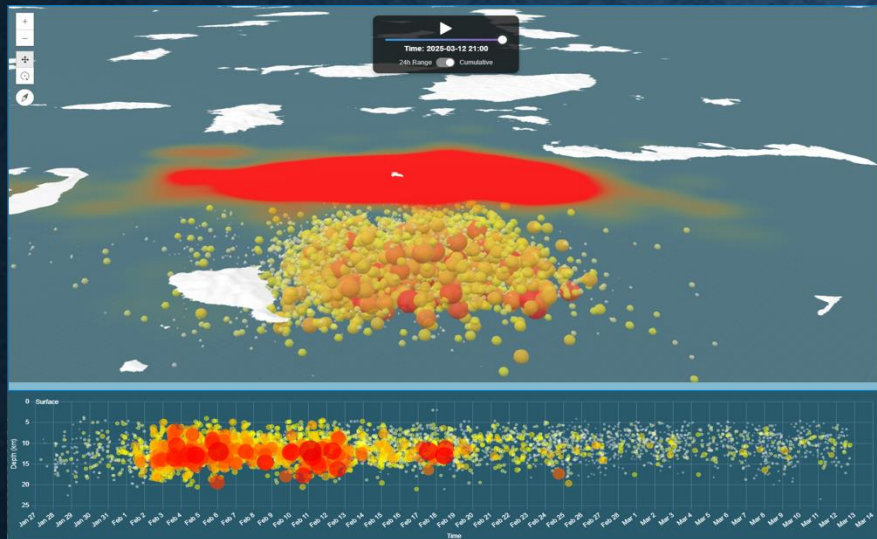


Research Proposal Support for Computing Resources:
Cyberinfrastructure and staffing costs, data management plans, ACCESS proposals



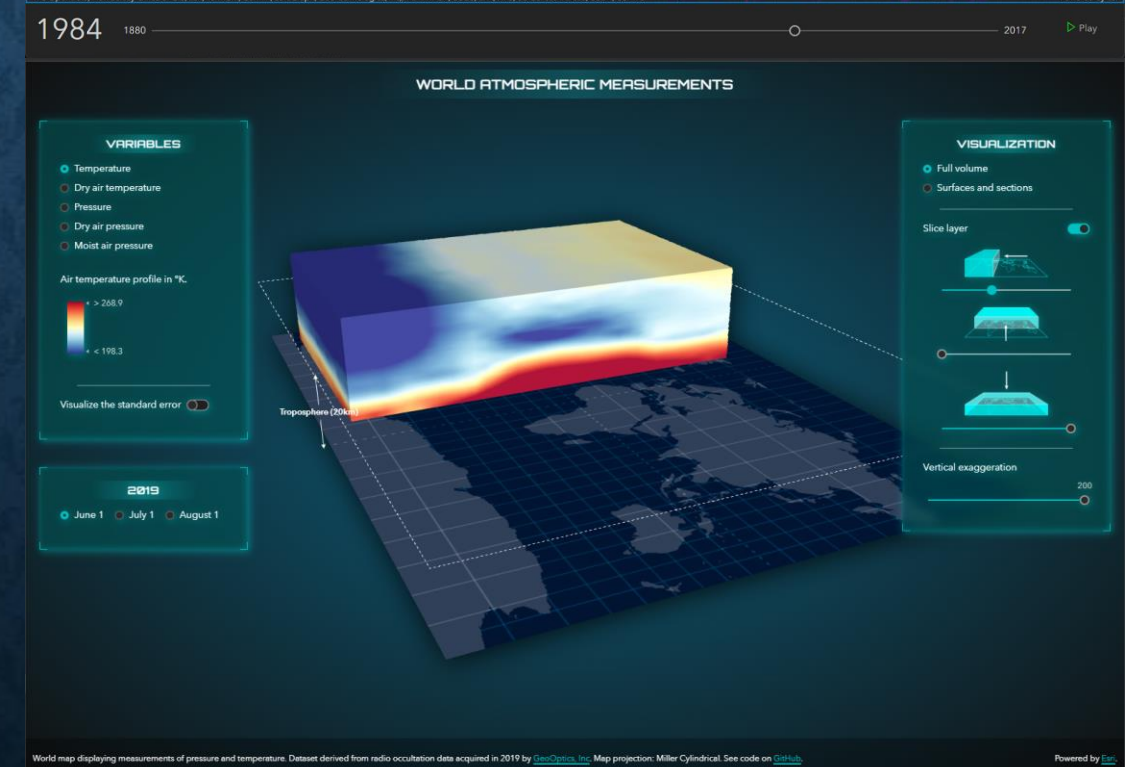
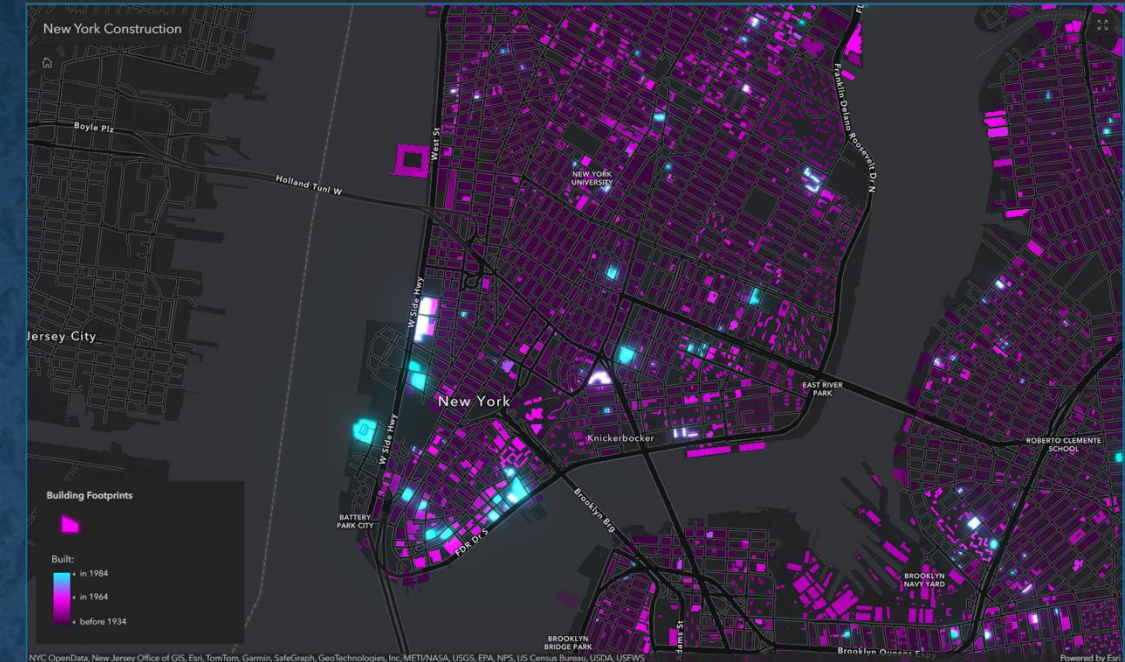
SPATIAL STUDIES LAB PROJECTS

- Cultural Heritage
- Disaster Response
- Sustainability and Resilience
- Public Health



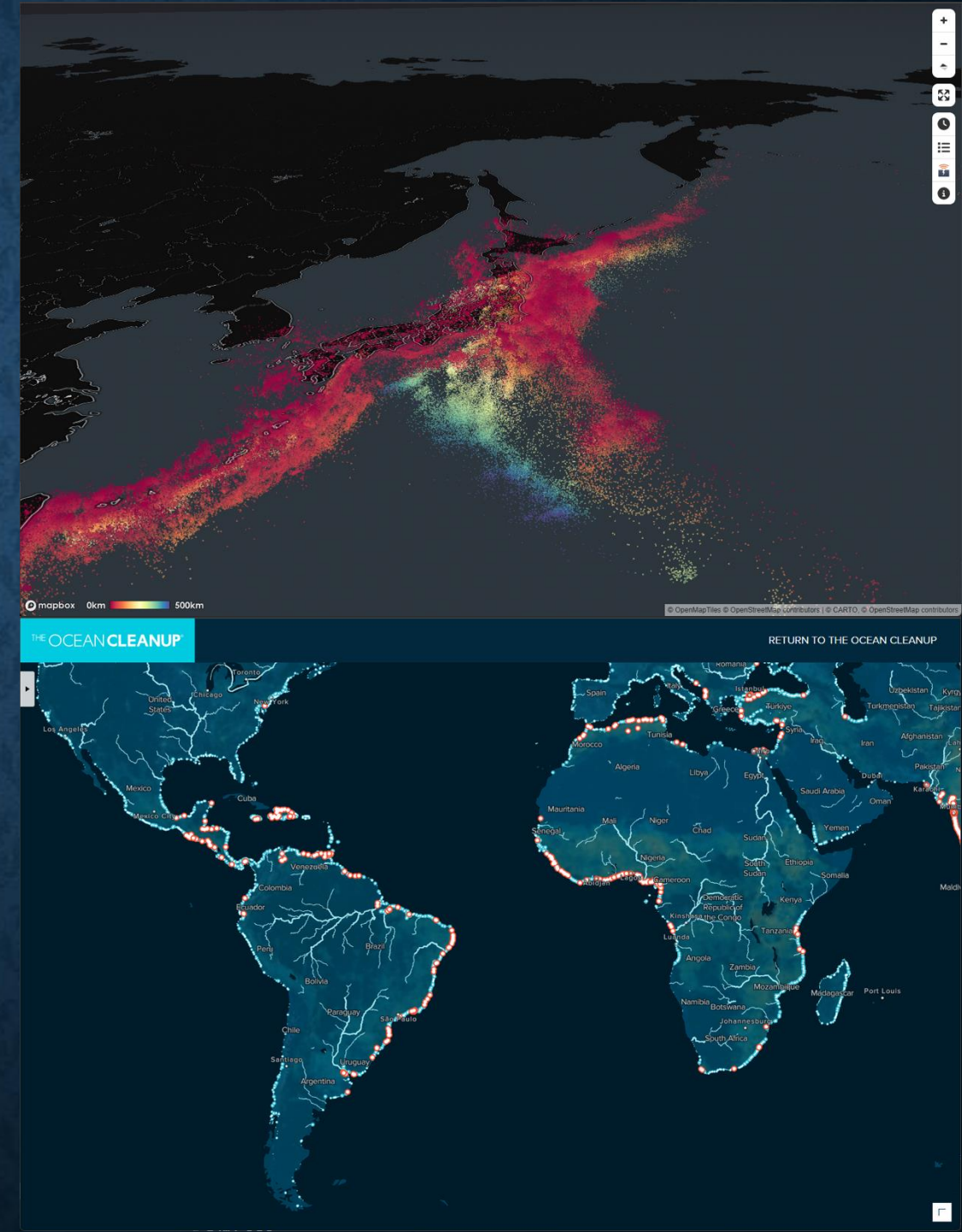
MAPPING & GEOSPATIAL APIS

- **ArcGIS JavaScript API**
 - Provides a full suite of mapping and GIS tools for building interactive maps and spatial applications.
 - Has support for visualization of large datasets.
 - Can display multidimensional scientific data through Voxel Layers.'



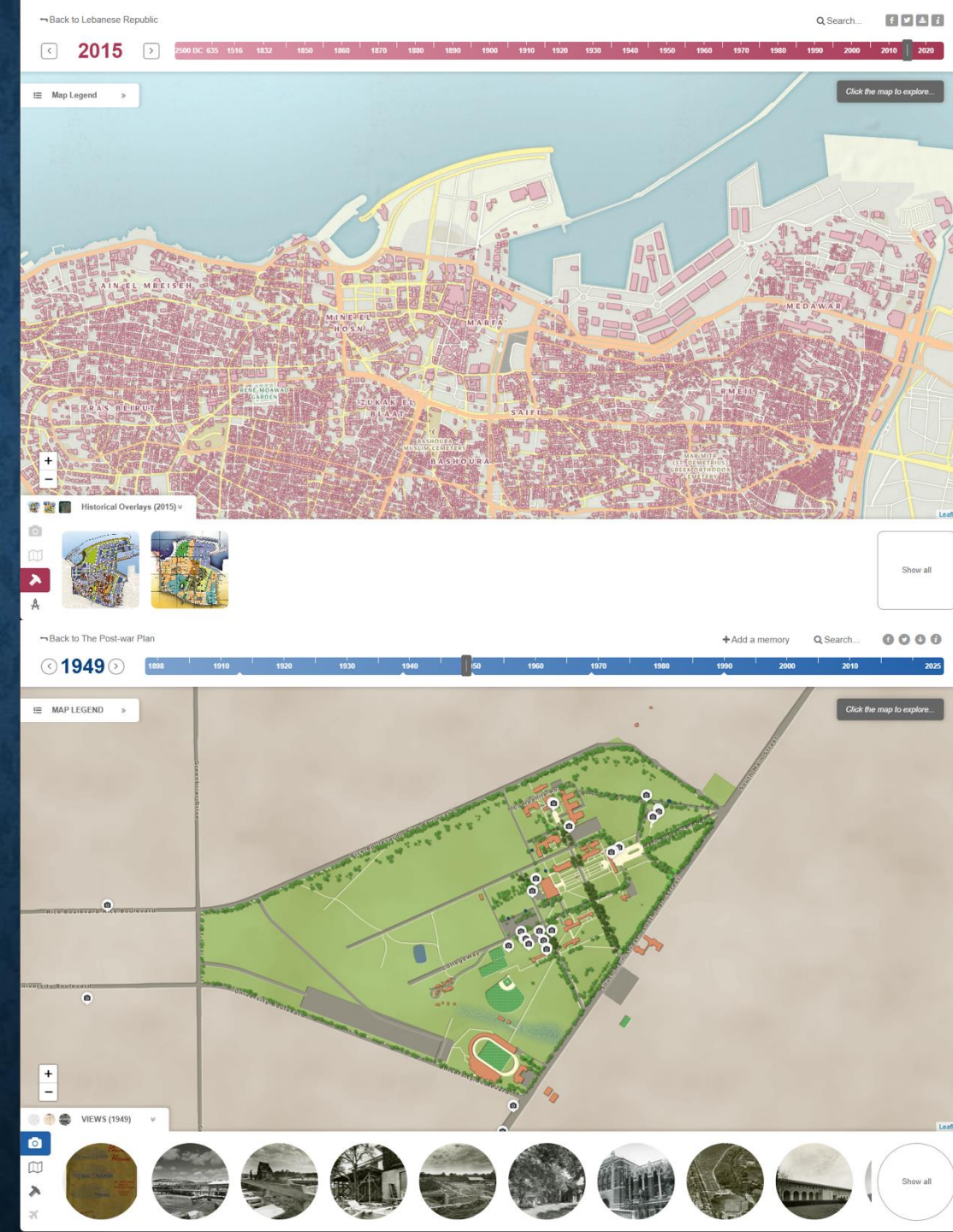
MAPPING & GEOSPATIAL APIS

- **Mapbox GL JS**
 - A platform that provides customizable, interactive maps and geospatial visualizations for web and mobile applications.
 - Has better performance than ArcGIS JS API



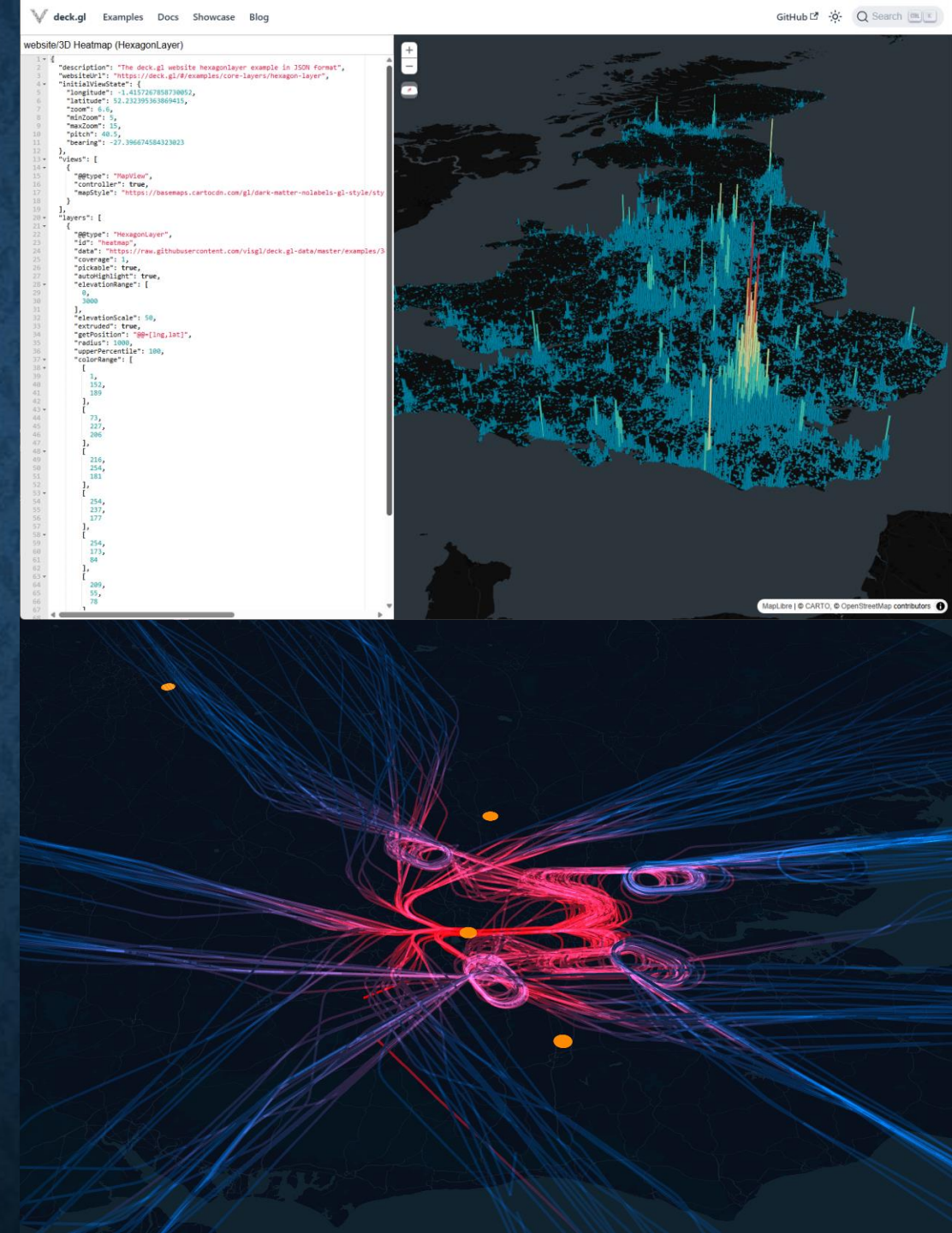
MAPPING & GEOSPATIAL APIS

- leaflet
 - an open-source JavaScript library for mobile-friendly interactive maps
 - Has large catalog of plugins



MAPPING & GEOSPATIAL APIS

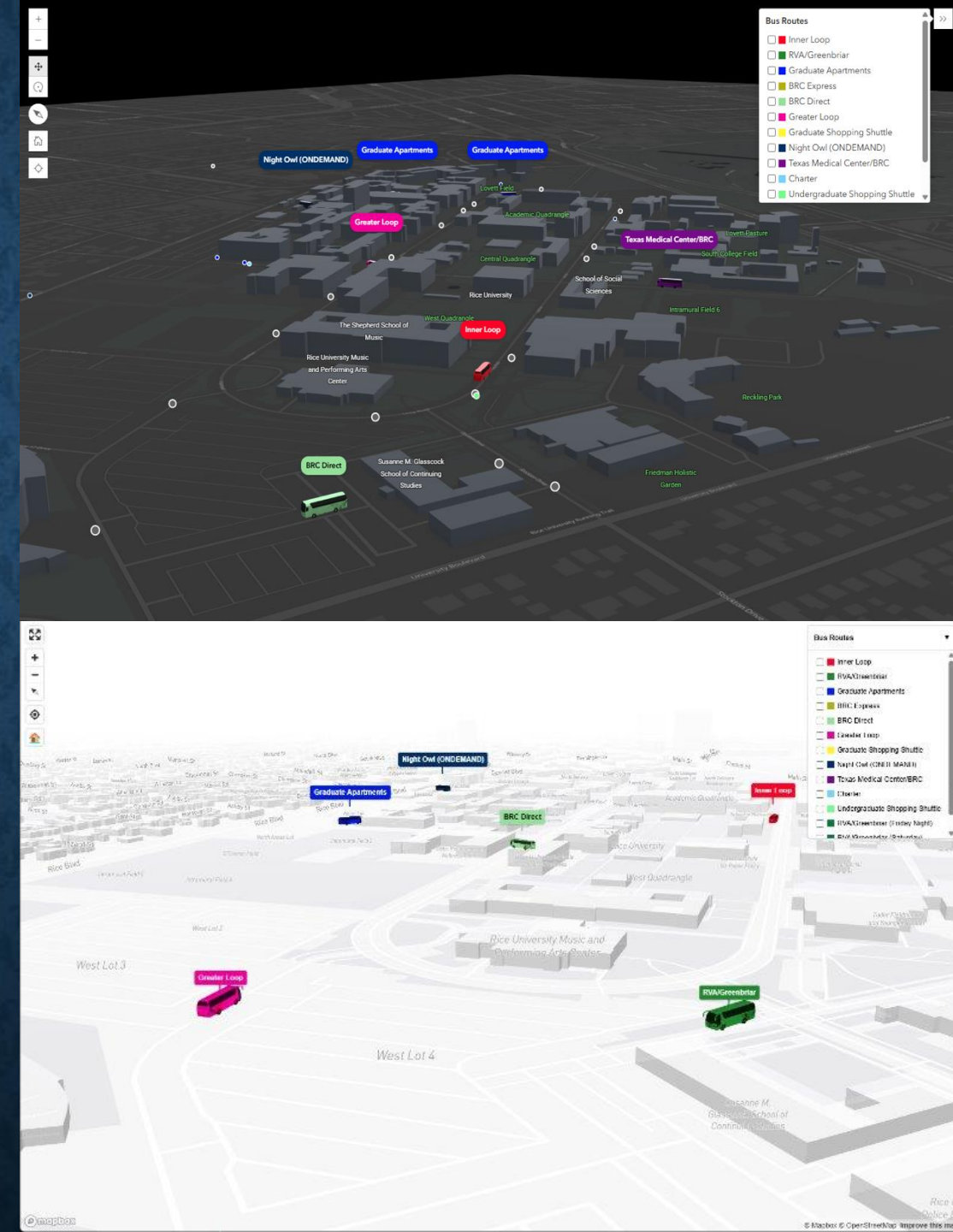
- **deck.gl**
 - GPU-powered framework for visual exploratory data analysis of large datasets.
 - Works with ArcGIS JS API, MapBox API and leafLet API, and more.



3D GRAPHICS LIBRARIES

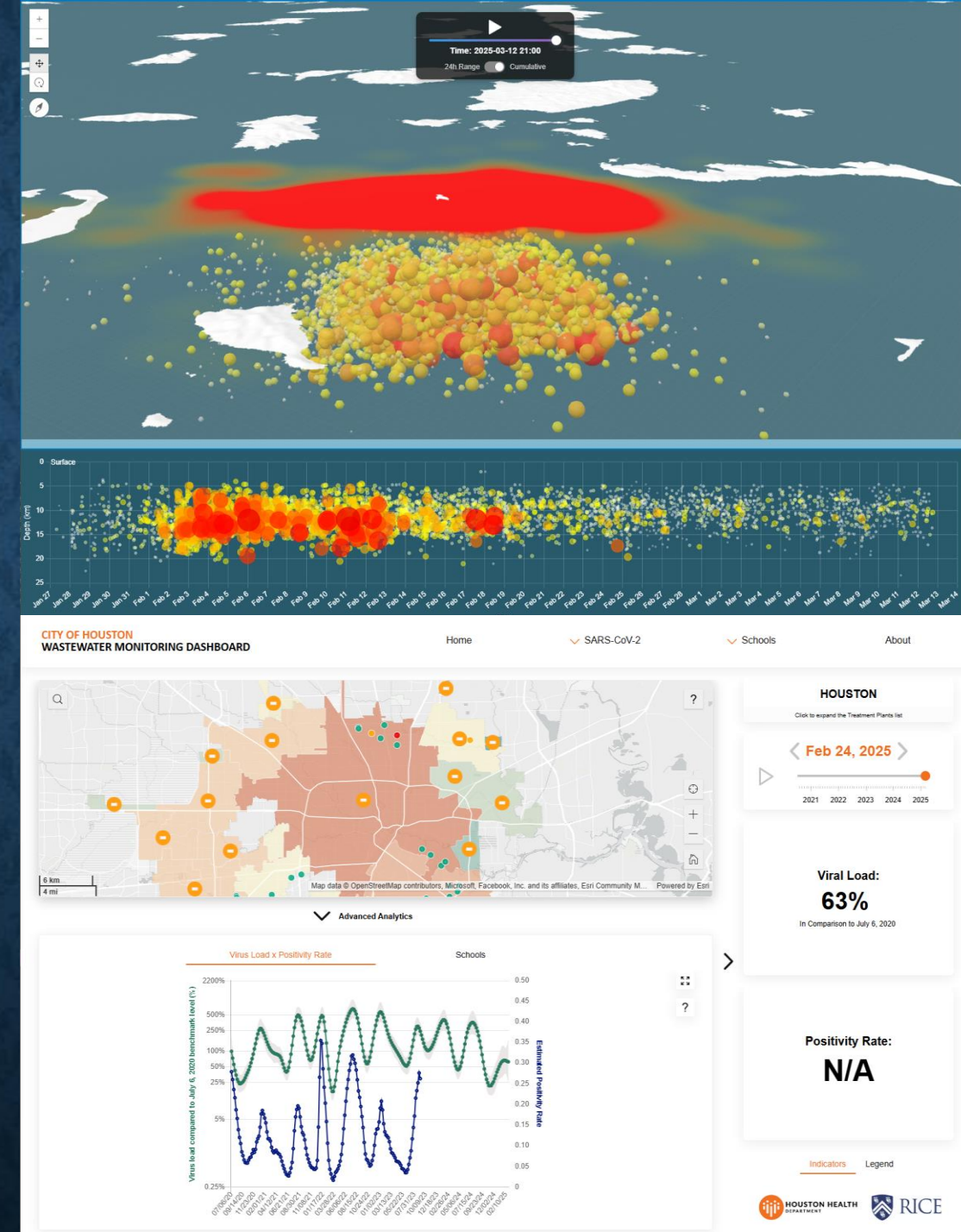
- **Three.js**

- cross-browser JavaScript library and API used to create and display animated 3D computer graphics in a web browser using WebGL
- Works with many Mapping APIs including ArcGIS Javascript API and Mapbox
- Similar libraries: Babylon.js



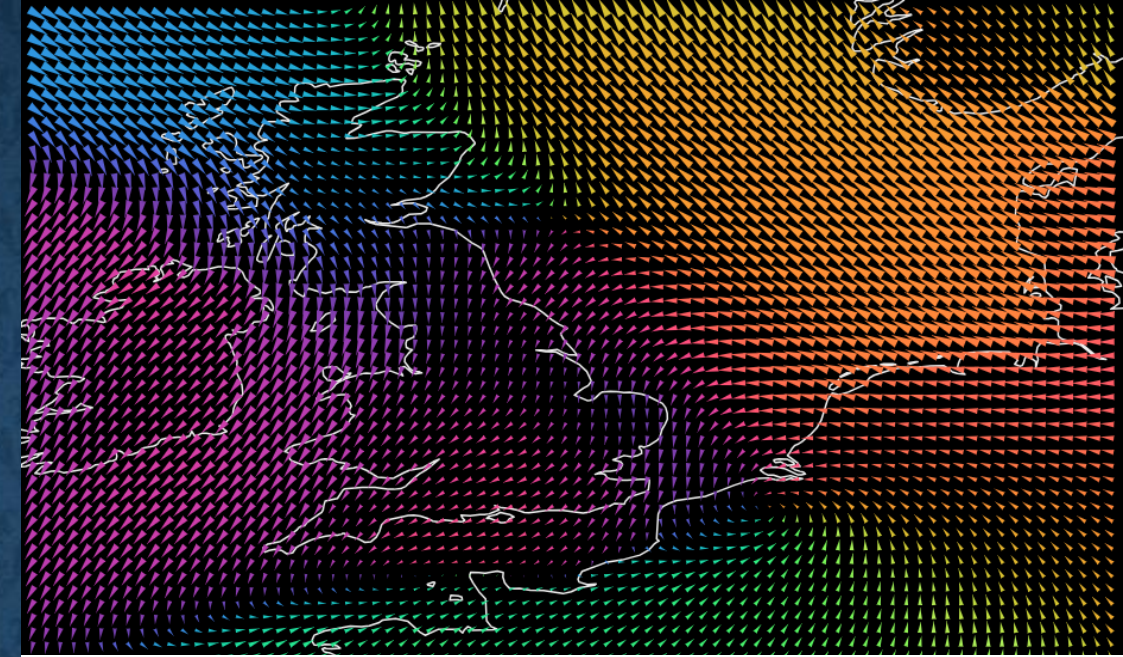
DATA VISUALIZATION LIBRARIES

- Chart.js
 - Simple yet flexible JavaScript charting library for the modern web



DATA VISUALIZATION LIBRARIES

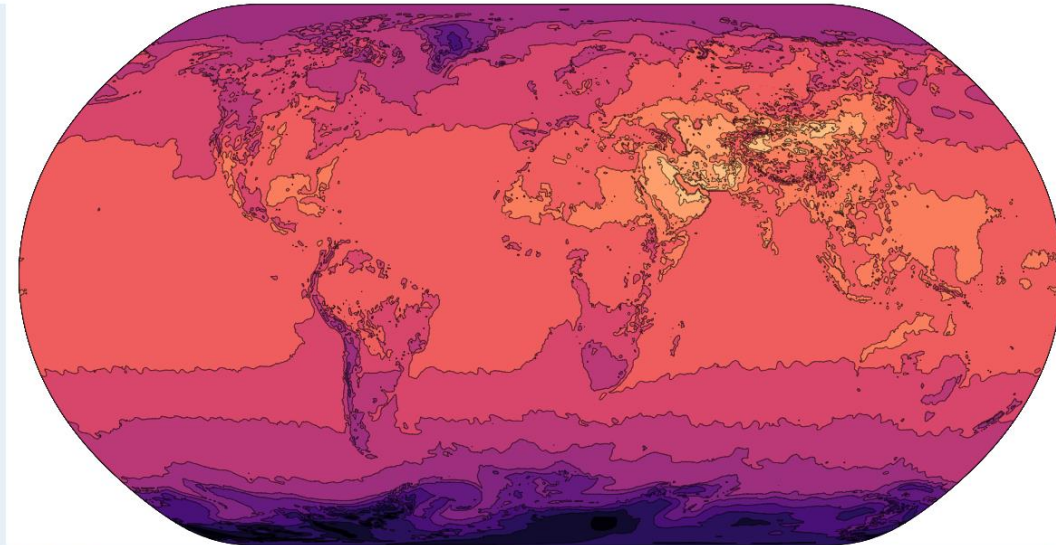
- D3.js
 - JavaScript library for producing dynamic, interactive data visualizations in web browsers



D3 › GALLERY

GeoTIFF contours II

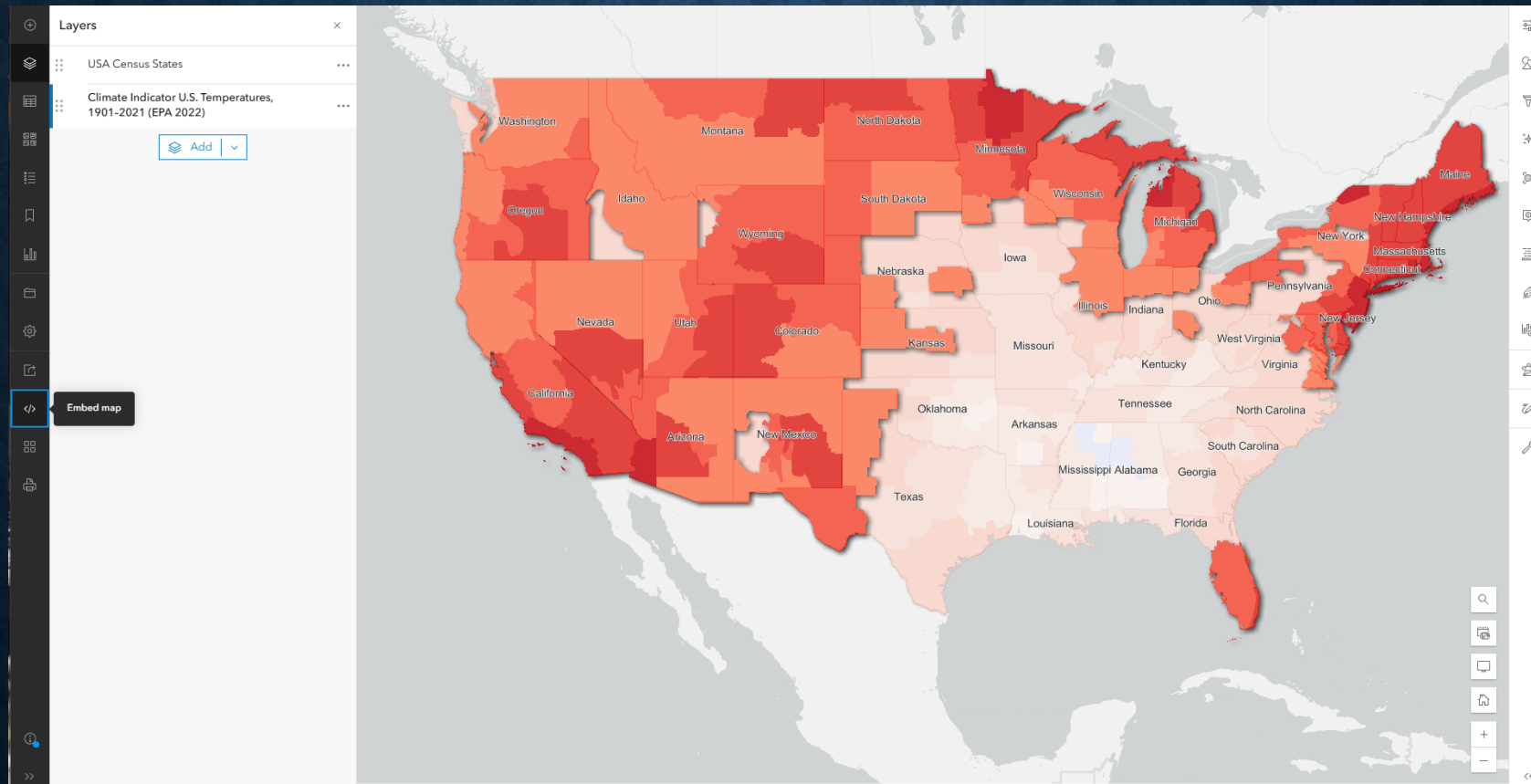
These contours are computed in equirectangular coordinates and then reprojected.



```
() chart = html`<svg style="width: 100%; height: auto; display: block;" viewBox="0 0 960 500">
  <g stroke="#000" stroke-width="0.5" stroke-linejoin="round" stroke-linecap="round">
    ${Array.from(contours(values), d => svg`<path d="${path(invert(d))}" fill="${color(d.value)}" />`)}
  </g>
</svg>
```


ARCGIS ONLINE

FIRST MAP



- Go to ricegis.maps.arcgis.com

RESOURCES

- **ArcGIS JavaScript API**

- [Overview and Examples](#)
- [Voxel Layers](#)

- **Mapbox GL JS**

- [Documentation](#)
- [Examples](#)

- **Leaflet JS**

- [Overview](#)
- [Tutorials](#)

- **Deck.gl**

- [Overview](#)
- [Examples](#)

- **Three.js**

- [Overview](#)
- [Documentation](#)

- **Babylon.js**

- [Overview](#)
- [Documentation](#)

- **Chart.js**

- [Overview](#)
- [Samples](#)

- **D3.js**

- [Overview](#)
- [Gallery](#)

THANK YOU

QUESTIONS?

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